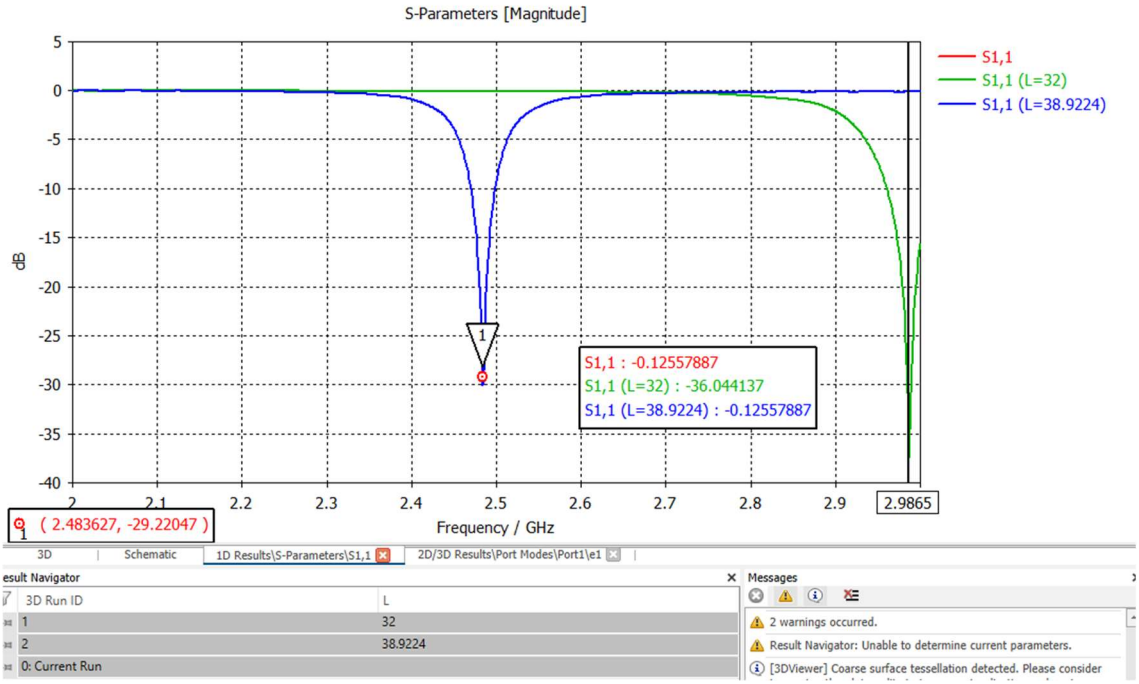
Sergi Salvà Pibernat
Eugeni Valentí Soler

Laboratory 3 Report WLA

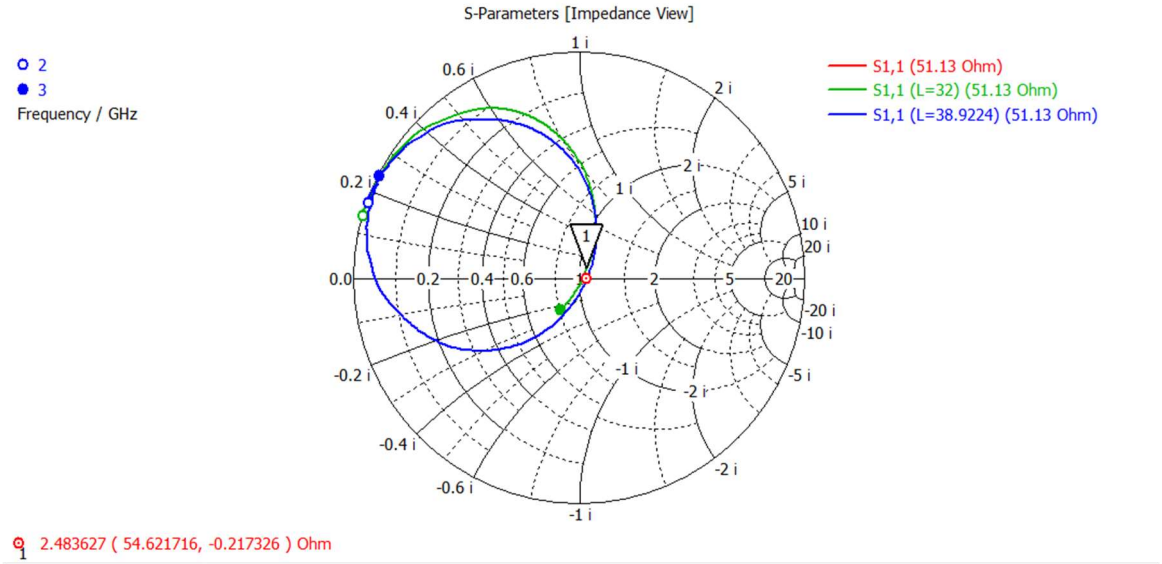
Exercise 1:

Following the tutorial, we obtained the following results:

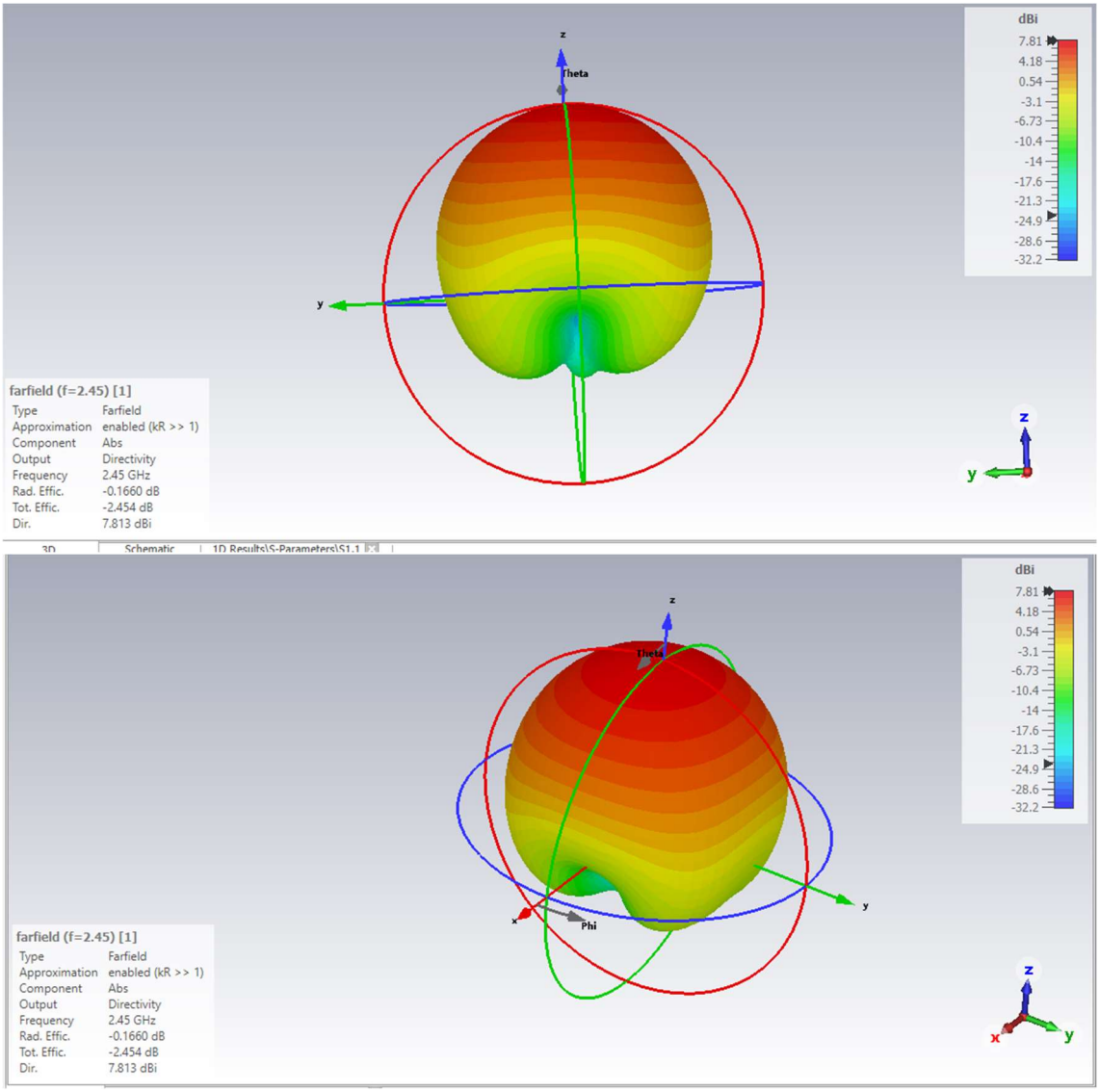
Here we can observe the amplitude of the S-Parameter in function of the Frequency. In green we see the first approach we made, and we obtained the minimum at 2.9865GHz instead of 2.45GHz. By adjusting the parameters correctly we obtained the minimum at a frequency of 2.48GHz as seen at the next figure.



On the following figure we can observe the smith chart of the simulation. We can see that we obtain an impedance of 54.62 Ohms.

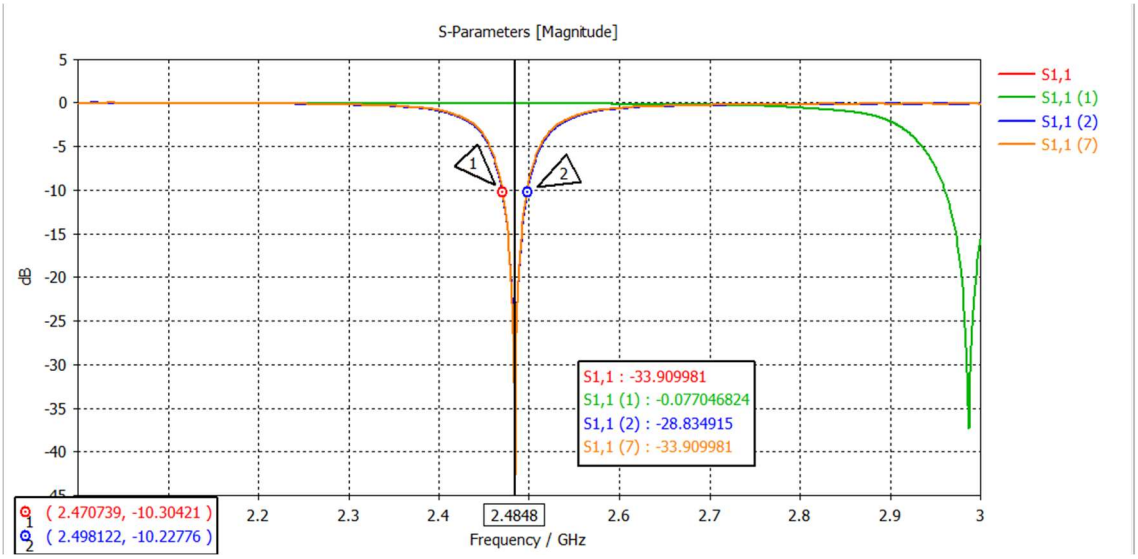


On the two following figures we can see the radiation pattern of the antenna:

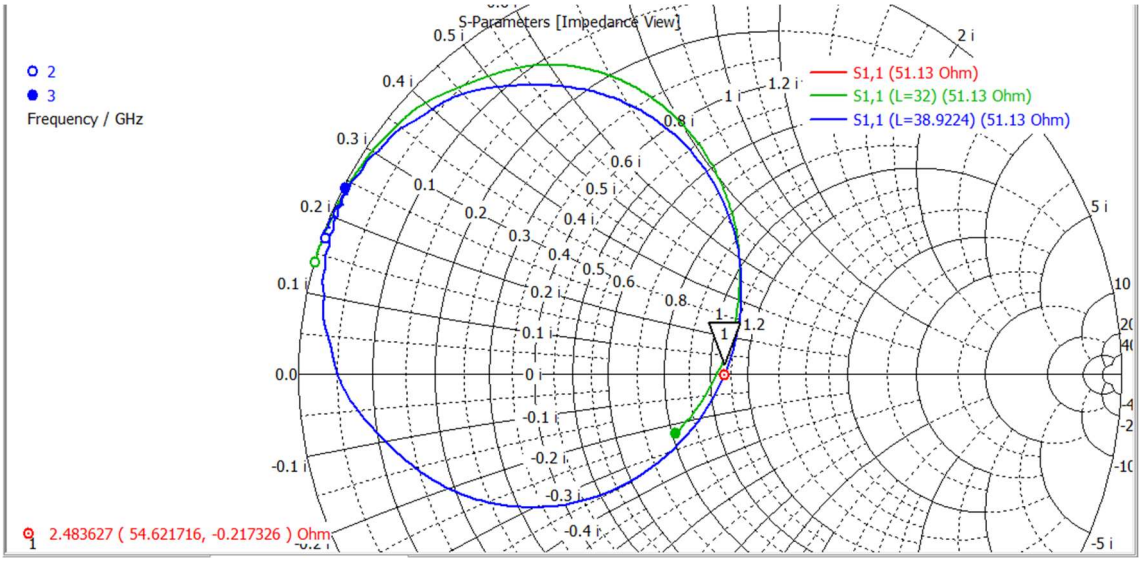


Exercise 2:

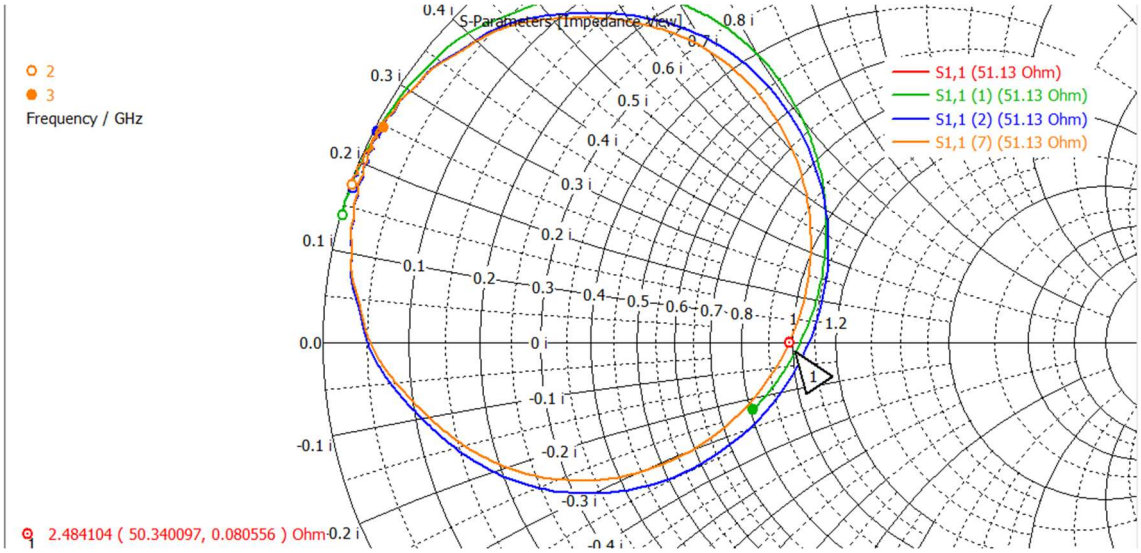
Following the instructions of exercise 2 we modified the patch antenna dimensions and we obtained the results of the following figures.



Here we can observe the smith chart plot of the new simulation and it can be observed that we obtain a 54.62Ohms impedance.

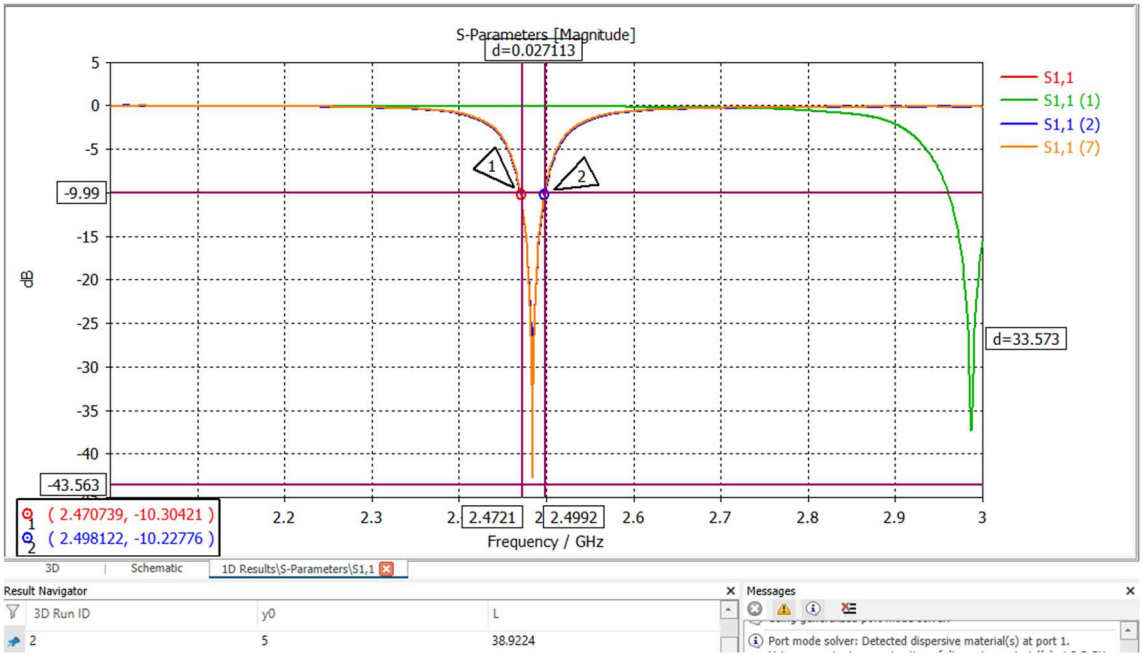


By modify the variable CoaxPos to obtain a 50Ohm impedance as it can be seen in the figure below:



To estimate the antenna Q we measure the Return Loss Fractional Bandwith around -10 dB. The FBW is 0.027113 GHz. Therefore, the estimated Q is:

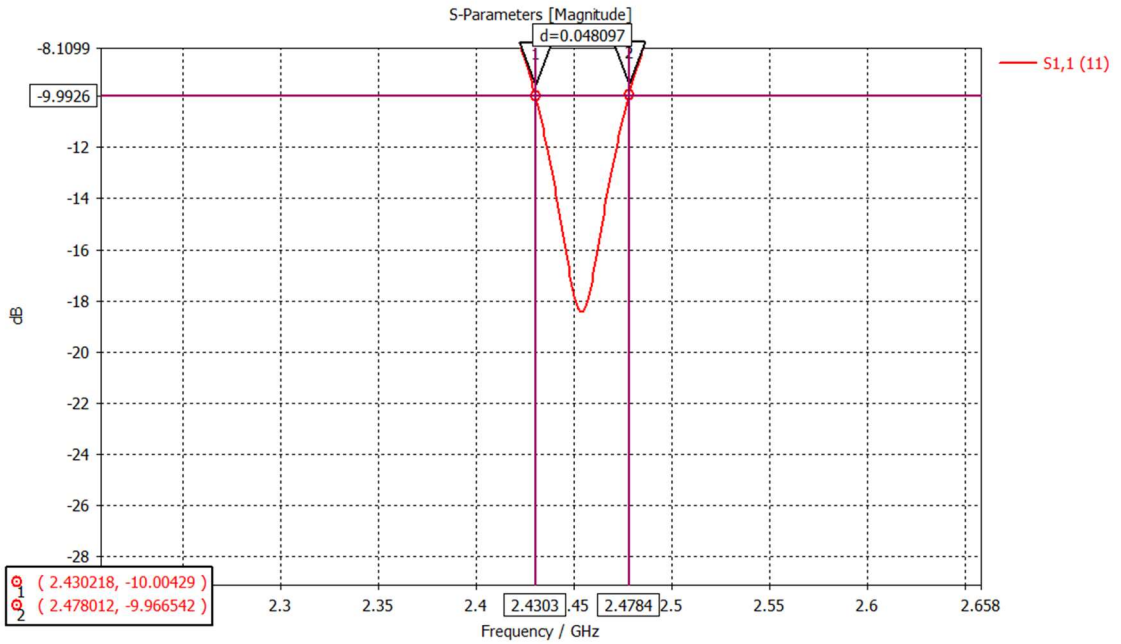
$$Q = \frac{0.707}{0.027113} = 26.06$$



Exercise 3:

Now, with the double height of the patch antenna we obtain the Return Loss Fractional Bandwith around -10 dB. The FBW is 0.048097 GHz. Therefore, the estimated Q is:

$$Q = \frac{0.707}{0.048097} = 14{,}7$$



As the height of the antenna increases the Q reduces and the bandwidth increases.